AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_9CC66FD754
Taxa	Triceratops
Clade	Ornithischian
PBDB Records	166
Mass (regression)	8247.9 kg (Campione & Evans 2012)
Mass (volumetric)	8350.1 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	8.790 m³
Validity Score	1.0000 / 1.0000
TCI	0.6411 (partial)
Axiom I Entropy	0.9501
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.4595
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.611, 1.742, 0.0]
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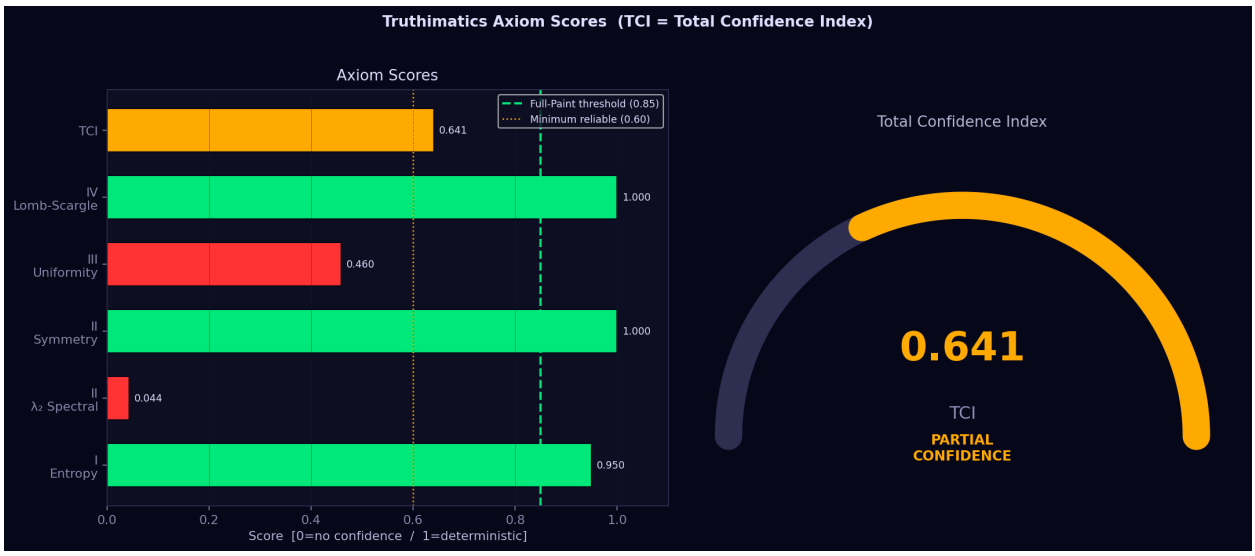
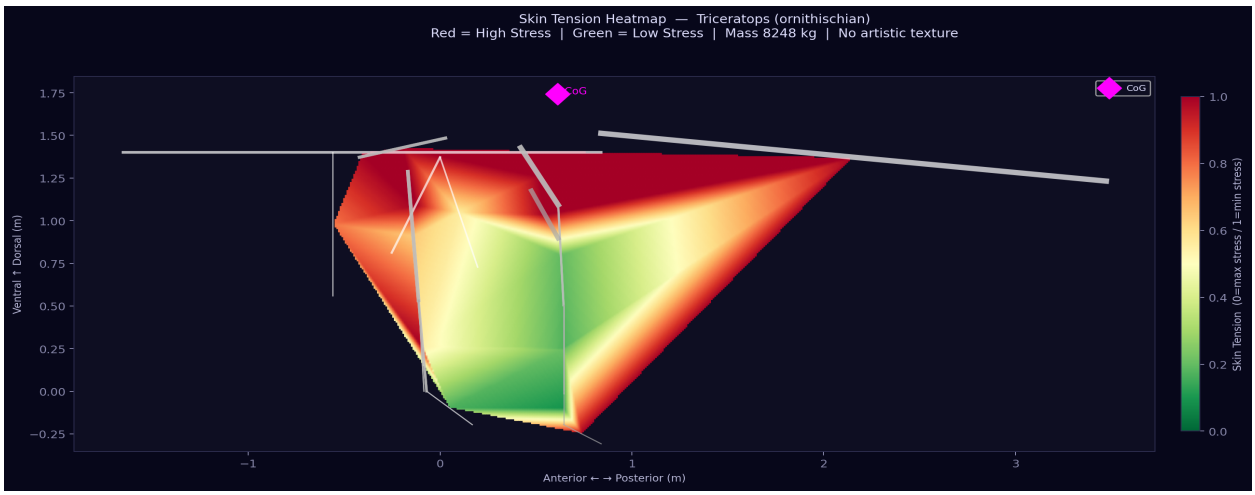
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



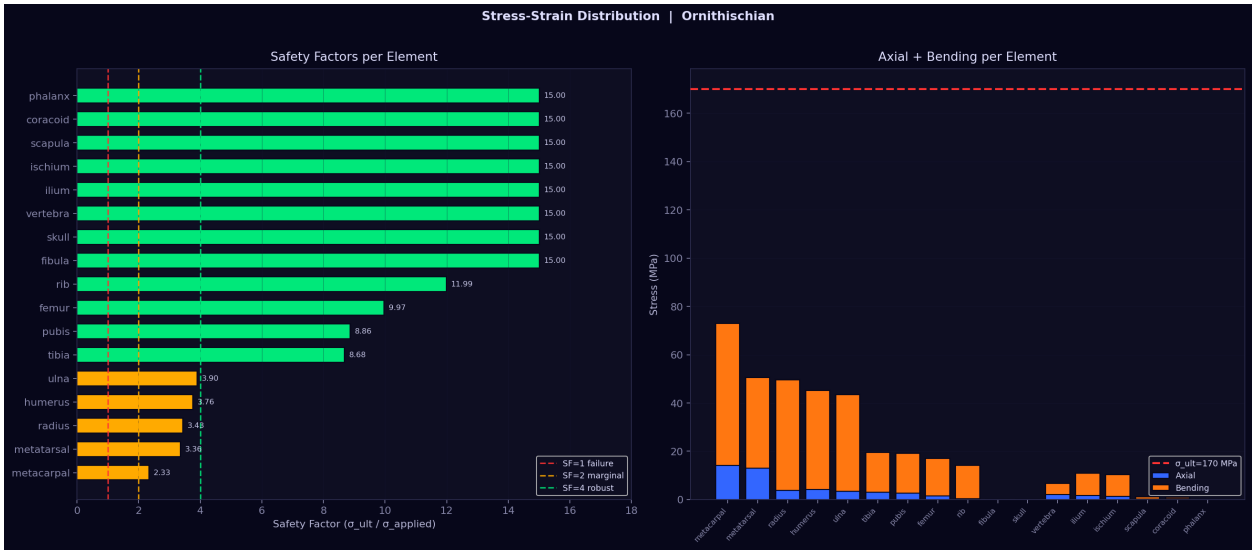
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

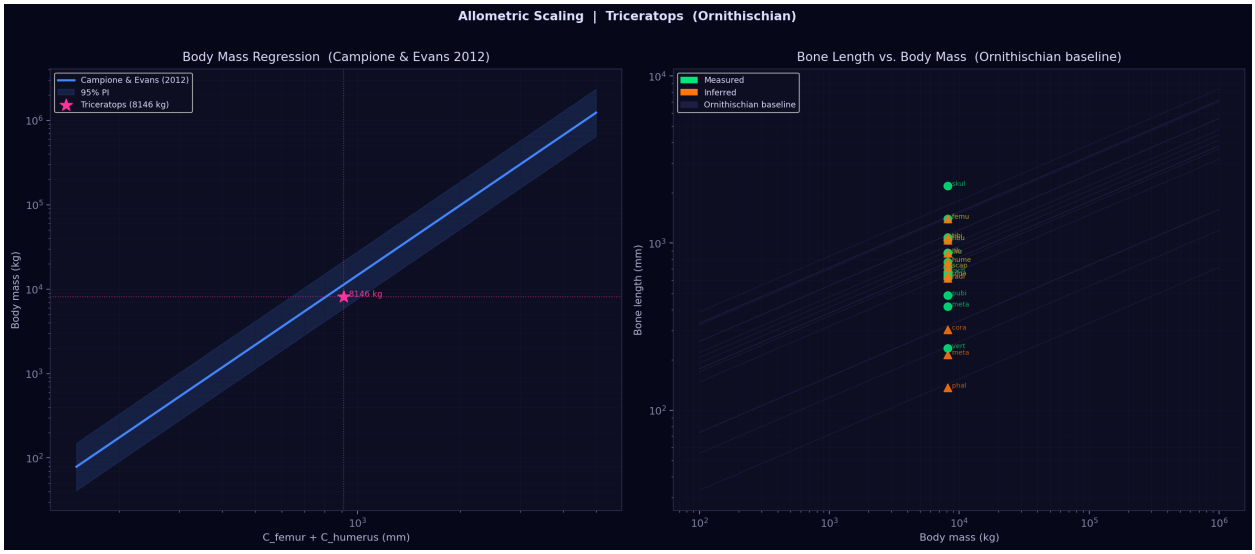


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.309	0.529	0.838	15.000	OK
femur	1.506	15.547	17.052	9.969	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	4.276	40.920	45.196	3.761	WARNING
ilium	1.686	9.136	10.822	15.000	OK
ischium	1.392	8.873	10.264	15.000	OK
metacarpal	14.216	58.698	72.914	2.332	WARNING
metatarsal	12.975	37.573	50.547	3.363	WARNING
phalanx	0.000	0.000	0.000	15.000	OK
pubis	2.596	16.590	19.186	8.860	OK
radius	3.775	45.829	49.604	3.427	WARNING
rib	0.429	13.753	14.182	11.987	OK
scapula	0.243	0.870	1.113	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	3.044	16.548	19.592	8.677	OK
ulna	3.367	40.176	43.544	3.904	WARNING
vertebra	2.159	4.480	6.638	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.898 m	Femur + tibia + metatarsal
Estimated Speed	17.1 km/h (4.75 m/s)	Gait: Trot | Fr=0.793
Stride Length	6.2 m	Alexander (1976) / Thulborn (1990)
Bite Force	193847.2 N (19760.2 kgf)	Ornithischian regression (Norman & Weishampel 1991)
Skull Length	2200.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.4084	Circumference / length (dimensionless)
Limb Strength Index	4.7e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	12.43 m	Allometric from femur length
Est. Neck Length	1.37 m	Allometric from femur length
Est. Tail Length	3.48 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	303.3	237.7	6863.52	8567.7	Constraint	0.40	15.00
femur	1400.0	182.0	18575.09	23187.3	Measured	0.88	9.97
femur_right	1400.0	182.0	18575.09	23187.3	Symmetry	0.84	9.97
fibula	1040.0	52.0	795.12	833.9	Measured	0.88	15.00
fibula_right	1040.0	52.0	795.12	833.9	Symmetry	0.84	15.00
humerus	770.0	108.0	3597.48	4490.7	Measured	0.88	3.76
humerus_right	770.0	108.0	3597.48	4490.7	Symmetry	0.84	3.76
ilium	868.0	172.0	10285.77	12839.7	Measured	0.88	15.00
ilium_right	868.0	172.0	10285.77	12839.7	Symmetry	0.84	15.00
ischium	665.0	112.0	3341.32	4171.0	Measured	0.88	15.00
metacarpal	214.5	57.4	105.51	86.8	Constraint	0.65	2.33
metatarsal	418.0	62.0	643.61	803.4	Measured	0.88	3.36
phalanx	136.5	47.5	37.21	28.8	Constraint	0.40	15.00
pubis	488.0	82.0	1314.34	1640.7	Measured	0.88	8.86
radius	615.0	68.0	1139.08	1421.9	Measured	0.88	3.43
radius_right	615.0	68.0	1139.08	1421.9	Symmetry	0.84	3.43
rib	875.0	44.0	678.54	847.0	Measured	0.88	11.99
scapula	715.0	268.0	20570.07	25677.6	Measured	0.88	15.00
scapula_right	715.0	268.0	20570.07	25677.6	Symmetry	0.84	15.00
skull	2200.0	820.0	592530.14	739655.4	Measured	0.88	15.00
tibia	1080.0	128.0	7087.67	8847.5	Measured	0.88	8.68
tibia_right	1080.0	128.0	7087.67	8847.5	Symmetry	0.84	8.68
ulna	640.0	72.0	1328.94	1658.9	Measured	0.88	3.90
ulna_right	640.0	72.0	1328.94	1658.9	Symmetry	0.84	3.90
vertebra	235.0	152.0	2174.78	2714.8	Measured	0.88	15.00